ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

At equilibrium the velocity should be zero. The torque and gravity torque must balance.

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state $\dot{\theta}$ = 0, and I assume $\theta_{ss} = \theta_d = 90°$ because the PD controller tracks the desired angle. Thus the steady-state error is zero.

Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

The Tustin coefficients are y[k]=0.881765y[k-1]+0.0591175(x[k]+x[k-1]). The DC gain is 0.0591175/(1-0.881765)=0.5.

Q3(c) [4 marks]

With $L=[\ell_1,\ell_2]^T$ and C=[1,0], A-LC=[[$-\ell_1$,1], [$-\ell_2$,-2]]. Its characteristic polynomial is det(sI-(A-LC)) = $(s+\ell_1)(s+2)+\ell_2 = s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. Desired poles at -10 give $(s+10)^2=s^2+20s+100$. Hence $\ell_1+2=20$ gives $\ell_1=18$, and $2\ell_1+\ell_2=100$ gives $\ell_2=64$.

Expanding $(s+\ell_1)(s+2)+\ell_2$ gives $s^2+(\ell_1+2)s+(2\ell_1+\ell_2)$. The desired polynomial is $(s+10)^2 = s^2+20s+100$. Matching the s coefficient gives $\ell_1 = 18$; matching the constant term then gives $36+\ell_2 = 100$, hence $\ell_2 = 64$.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) [2 marks]

Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) [2 marks]

Integral action removes steady-state error, although this only holds for the nominal values of b and c. Saturation can cause windup, which should be handled by anti-windup.